Pattern Wise DSA: Click Here

DSA GUIDE

DAY 1 - 5 in 1st Week

(Mandatory Things to know in Any Language And Build Up Logical Thinking)

Subtopic	Time	Video Resource	Article Resource	Practice (LeetCode /Other)
User Input / Output	30 mins	Youtube Link	Article	Practice Link
Data Types	30 mins	Youtube Link	Article	Practice Link
If Else Statements	40 mins	Youtube Link	Article	Practice Link
Switch Statement	40 mins	Youtube Link	Article	Practice Link
What are Arrays, Strings?	1 hr 20 mins	Youtube Link For Array Youtube Link For String	Article	Practice Question Given Below
For Loops	40 mins	Youtube Link	Article	Practice Question Given Below
While Loops And Do While Loops	40 mins	Youtube Link	Article	Practice Question Given Below
Functions/Methods (Pass by Reference and Value)	2 hr	Youtube Link	Article	Practice with Youtube Link
Time Complexity (Basics)	1 hr 20 mins	Youtube Link	Article	-----------

Practice Question For Arrays and Strings

✅ Array Questions (If-Else Focused)

1. **Positive or Negative**
   - **Problem:** Given a number, check if it is positive, negative, or zero.
   - **Example:** Input: `-5` → Output: `Negative`

2. **Even or Odd**
   - **Problem:** Check if a number is even or odd using if-else.
   - **Example:** Input: `7` → Output: `Odd`

3. **Compare Two Numbers**
   - **Problem:** Take two numbers as input and print which one is greater, or if they are equal.
   - **Example:** Input: `4, 4` → Output: `Equal`

✅ String Questions (If-Else Focused)

1. **Check Empty String**
   - **Problem:** Given a string, check if it is empty or not.
   - **Example:** Input: `""` → Output: `Empty String`

2. **Check First Character**
   - **Problem:** Given a string, check if its first character is a vowel or consonant.
   - **Example:** Input: `"Apple"` → Output: `Vowel`

3. **String Length Check**
   - **Problem:** Given a string, check if its length is greater than 5.

    - **Example:** Input: `"Hi"` → Output: `Length is less than or equal to 5`

🚀 How to Practice:

1. Write each as a **small program with if-else only**.
2. Focus on **taking input, applying conditions, and printing output**.

Practice Question For Loops

✅ 5 Normal Loop Questions

1. **Print 1 to N**
    - **Problem:** Take an integer N and print numbers from 1 to N using a for loop.
2. **Sum of N Numbers**
    - **Problem:** Find the sum of all numbers from 1 to N using a while loop.
3. **Factorial of a Number**
    - **Problem:** Calculate factorial of a given number using a for loop.
4. **Reverse a Number**
    - **Problem:** Take an integer input and reverse its digits using a while loop.
    - **Example:** Input: `1234` → Output: `4321`
5. **Count Digits**
    - **Problem:** Count the number of digits in a given number using a do-while loop.

✅ 5 Pattern Questions

Print Solid Rectangle

```
*****
*****
*****
```

Print Hollow Rectangle

```
*****
*   *
*****
```

Print Half Pyramid

```
*
**
***
****
```

Print Inverted Half Pyramid

```
****
***
**
*
```

Print Numbers Pyramid

```
1
12
123
1234
```

Solve any pattern Question - [Youtube Link]() and Practice More pattern for logic building.

🚀 How to Practice Today

✔️ Try each with a for **loop, while loop, and do-while loop** interchangeably for mastery.
✔️ Focus on **logic clarity and neat output formatting**.

DAY 6 - 7 in 1st Week

Reverse String - Practice Link Leetcode

Check Palindrome - Practice Link CodingNinja

Valid Palindrome - Practice Link Leetcode

Reverse words - Practice Link GFG

Maximum Occurring Character - Practice Link GFG

Replace Spaces - Practice Link CodingNinja

Remove All Occurrences of SubString - Practice Link Leetcode

String Compression - Practice link Leetcode

Permutation In String - Practice Link Leetcode

Remove All Adjacent Duplicates In String - Practice Link Leetcode

DAY 8 in 2nd Week

(Learn STL/Java Collection or Similar Thing in Your language)

Subtopic	Time	Video Resource	Article Resource	Practice (LeetCode /Other)
C++ STL	4 Hours	Youtube Link	Article	Practice With Youtube Video
Java Collections	4 Hours	Youtube Link	Article	Practice With Youtube Video

DAY 9-10 in 2nd Week

(Learn Basic Maths)

Subtopic	Time	Video Resource	Article Resource	Practice (LeetCode /Other)
Count Digits	1 Hour	Youtube Link	Article	Practice Link
Reverse a Number	1 Hour	Youtube Link	Article	Practice Link
Check Palindrome	1 Hour	Youtube Link	Article	Practice Link
GCD or HCF	1 Hour	Youtube Link	Article	Practice Link
Armstrong Numbers	1 Hour	Youtube Link	Article	Practice Link
Print all Divisors	1 Hour	Youtube Link	Article	Practice Link
Check for Prime	1 Hour	Youtube Link	Article	Practice Link

DAY 11-13 in 2nd Week

(Learn Basic Recursion)

Subtopic	Time	Video Resource	Article Resource	Practice (LeetCode/Other)
Understand recursion by printing something N times	2 Hour	Youtube Link	Article	Practice Link
Print name N times using recursion	30 Mins	Youtube Link	Article	Practice Link
Print 1 to N using recursion	30 Mins	Youtube Link	Article	Practice Link
Print N to 1 using recursion	30 Mins	Youtube Link	Article	Practice Link

Sum of first N numbers	30 Mins	Youtube Link	Article	Practice Link
Factorial of N numbers	30 Mins	Youtube Link	Article	Practice Link
Reverse an array	30 Mins	Youtube Link	Article	Practice Link
Check if a string is palindrome or not	30 Mins	Youtube Link	Article	Practice Link
Fibonacci Number	1 Hour	Youtube Link	Article	Practice Link

DAY 14-15 in 3rd Week

(Learn Basic Hashing)

Subtopic	Time	Video Resource	Article Resource	Practice (LeetCode/Other)
Hashing Theory	2 Hour	Youtube Link	Article	Practice with Youtube Video
Counting frequencies of array elements	2 Hour	Youtube Link	Article	Practice Link
Find the highest/lowest frequency element	2 Hour	Youtube Link	Article	Practice Link

DAY 16-20 in 3rd Week

(Learn Sorting)

Subtopic	Time	Video Resource	Article Resource	Practice (LeetCode/Other)
Selection Sort	2 Hours	Youtube Link	Article	Practice Link
Bubble Sort	2 Hours	Youtube Link	Article	Practice Link
Insertion Sort	2 Hours	Youtube Link	Article	Practice Link
Merge Sort	2 Hours	Youtube Link	Article	Practice Link
Recursive Bubble Sort +Selection Sort	2 Hours	Youtube Link	Article	Practice Link
Recursive Insertion Sort	1 Hour	Youtube Link	Article	Practice Link
Quick Sort	2 Hour	Youtube Link	Article	Practice Link

DAY 21-34 in 4Th and 5Th Week

(Learn Array)

Do First Full PlayList of Array Array Playlist
Then Start to solve these Questions..

Array 1	Reverse the array
Array 2	Find the maximum and minimum element in an array
Array 3	Find the "Kth" max and min element of an array
Array 4	Given an array which consists of only 0, 1 and 2. Sort the array without using any sorting algo
Array 5	Move all the negative elements to one side of the array
Array 6	Find the Union and Intersection of the two sorted arrays.
Array 7	Write a program to cyclically rotate an array by one.
Array 8	find Largest sum contiguous Subarray [V. IMP]
Array 9	Minimise the maximum difference between heights [V.IMP]
Array 10	Minimum no. of Jumps to reach end of an array

Array 11	find duplicate in an array of N+1 Integers
Array 12	Merge 2 sorted arrays without using Extra space.
Array 13	Kadane's Algo [V.V.V.V.V IMP]
Array 14	Merge Intervals
Array 15	Next Permutation
Array 16	Count Inversion
Array 17	Best time to buy and Sell stock
Array 18	find all pairs on integer array whose sum is equal to given number
Array 19	find common elements In 3 sorted arrays
Array 20	Rearrange the array in alternating positive and negative items with O(1) extra space
Array 21	Find if there is any subarray with sum equal to 0
Array 22	Find factorial of a large number
Array 23	find maximum product subarray
Array 24	Find longest consecutive subsequence
Array 25	Given an array of size n and a number k, fin all elements that appear more than " n/k " times.
Array 26	Maximum profit by buying and selling a share at most twice
Array 27	Find whether an array is a subset of another array
Array 28	Find the triplet that sum to a given value
Array 29	Trapping Rain water problem
Array 30	Chocolate Distribution problem
Array 31	Smallest Subarray with sum greater than a given value
Array 32	Three way partitioning of an array around a given value
Array 33	Minimum swaps required bring elements less equal K together
Array 34	Minimum no. of operations required to make an array palindrome
Array 35	Median of 2 sorted arrays of equal size
Array 36	Median of 2 sorted arrays of different size

DAY 35-39 in 6Th Week

(Learn Matrix)

Do First Matrix Playlist
Then Start to solve these Questions..

Matrix 1	Spiral traversal on a Matrix
Matrix 2	Search an element in a matrix
Matrix 3	Find median in a row wise sorted matrix
Matrix 4	Find row with maximum no. of 1's
Matrix 5	Print elements in sorted order using row-column wise sorted matrix
Matrix 6	Maximum size rectangle
Matrix 7	Find a specific pair in matrix
Matrix 8	Rotate matrix by 90 degrees
Matrix 9	Kth smallest element in a row-column wise sorted matrix
Matrix 10	Common elements in all rows of a given matrix

DAY 40-49 in 7Th Week

(Learn String)

Do First Strings & StringBuilder Playlist
Then Start to solve these Questions..
✅ You have **9 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 9 days**, you will do later **when you practice** (during your future revision or practice sessions).

String 1	Reverse a String
String 2	Check whether a String is Palindrome or not
String 3	Find Duplicate characters in a string
String 4	Why are strings immutable in Java?
String 5	Write a Code to check whether one string is a rotation of another

String 6	Write a Program to check whether a string is a valid shuffle of two strings or not
String 7	Count and Say problem
String 8	Write a program to find the longest Palindrome in a string.[Longest palindromic Substring]
String 9	Find Longest Recurring Subsequence in String
String 10	Print all Subsequences of a string.
String 11	Print all the permutations of the given string
String 12	Split the Binary string into two substring with equal 0's and 1's
String 13	Word Wrap Problem [VERY IMP].
String 14	EDIT Distance [Very Imp]
String 15	Find next greater number with same set of digits. [Very Very IMP]
String 16	Balanced Parenthesis problem.[Imp]
String 17	Word break Problem[Very Imp]
String 18	Rabin Karp Algo
String 19	KMP Algo
String 20	Convert a Sentence into its equivalent mobile numeric keypad sequence.
String 21	Minimum number of bracket reversals needed to make an expression balanced.
String 22	Count All Palindromic Subsequences in a given String.
String 23	Count of number of given string in 2D character array
String 24	Search a Word in a 2D Grid of characters.
String 25	Boyer Moore Algorithm for Pattern Searching.
String 26	Converting Roman Numerals to Decimal
String 27	Longest Common Prefix
String 28	Number of flips to make binary string alternate
String 29	Find the first repeated word in the string.
String 30	Minimum number of swaps for bracket balancing.
String 31	Find the longest common subsequence between two strings.
String 32	Program to generate all possible valid IP addresses from given string.

String 33	Write a program to find the smallest window that contains all characters of string itself.
String 34	Rearrange characters in a string such that no two adjacent are same
String 35	Minimum characters to be added at front to make string palindrome
String 36	Given a sequence of words, print all anagrams together
String 37	Find the smallest window in a string containing all characters of another string
String 38	Recursively remove all adjacent duplicates
String 40	String matching where one string contains wildcard characters
String 41	Function to find Number of customers who could not get a computer
String 42	Transform One String to Another using Minimum Number of Given Operation
String 43	Check if two given strings are isomorphic to each other
String 44	Recursively print all sentences that can be formed from list of word lists

DAY 50-63 in 8Th & 9Th Week

(Learn Searching and Sorting)

Do First Searching and Sorting
We already completed **Sorting** during **Days 16–20 of Week 3**.
Now, let's **focus on Searching** and start solving questions.
✅ You have **13 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 13 days**, you will do later **when you practice** (during your future revision or practice sessions).

Searching & Sorting 1	Find first and last positions of an element in a sorted array
Searching & Sorting 2	Find a Fixed Point (Value equal to index) in a given array

Searching & Sorting 3	Search in a rotated sorted array
Searching & Sorting 4	square root of an integer
Searching & Sorting 5	Maximum and minimum of an array using minimum number of comparisons
Searching & Sorting 6	Optimum location of point to minimize total distance
Searching & Sorting 7	Find the repeating and the missing
Searching & Sorting 8	find majority element
Searching & Sorting 9	Searching in an array where adjacent differ by at most k
Searching & Sorting 10	find a pair with a given difference
Searching & Sorting 11	find four elements that sum to a given value
Searching & Sorting 12	maximum sum such that no 2 elements are adjacent
Searching & Sorting 13	Count triplet with sum smaller than a given value
Searching & Sorting 14	merge 2 sorted arrays
Searching & Sorting 15	print all subarrays with 0 sum
Searching & Sorting 16	Product array Puzzle
Searching & Sorting 17	Sort array according to count of set bits
Searching & Sorting 18	minimum no. of swaps required to sort the array
Searching & Sorting 19	Bishu and Soldiers
Searching & Sorting 20	Rasta and Kheshtak
Searching & Sorting 21	Kth smallest number again
Searching & Sorting 22	Find pivot element in a sorted array
Searching & Sorting 23	K-th Element of Two Sorted Arrays

Searching & Sorting 24	Aggressive cows
Searching & Sorting 25	Book Allocation Problem
Searching & Sorting 26	EKOSPOJ:
Searching & Sorting 27	Job Scheduling Algo
Searching & Sorting 28	Missing Number in AP
Searching & Sorting 29	Smallest number with at least trailing zeroes infactorial
Searching & Sorting 30	Painters Partition Problem:
Searching & Sorting 31	ROTI-Prata SPOJ
Searching & Sorting 32	DoubleHelix SPOJ
Searching & Sorting 33	Subset Sums
Searching & Sorting 34	Find the inversion count
Searching & Sorting 35	Implement Merge-sort in-place
Searching & Sorting 36	Partitioning and Sorting Arrays with Many Repeated Entries

DAY 64-77 in 10Th & 11Th Week

(Learn LinkedList)

Do First LinkedList
Then Start to solve these Questions..
✅ You have **13 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 13 days**, you will do later **when you practice** (during your future revision or practice sessions).

LinkedList 1	Write a Program to reverse the Linked List. (Both Iterative and recursive)
LinkedList 2	Reverse a Linked List in a group of Given Size. [Very Imp]

LinkedList 3	Write a program to detect a loop in a linked list.
LinkedList 4	Write a program to Delete loop in a linked list.
LinkedList 5	Find the starting point of the loop.
LinkedList 6	Remove Duplicates in a sorted Linked List.
LinkedList 7	Remove Duplicates in a Un-sorted Linked List.
LinkedList 8	Write a Program to Move the last element to Front in a Linked List.
LinkedList 9	Add “1” to a number represented as a Linked List.
LinkedList 10	Add two numbers represented by linked lists.
LinkedList 11	Intersection of two Sorted Linked List.
LinkedList 12	Intersection Point of two Linked Lists.
LinkedList 13	Merge Sort For Linked lists.[Very Important]
LinkedList 14	Quicksort for Linked Lists.[Very Important]
LinkedList 15	Find the middle Element of a linked list.
LinkedList 16	Check if a linked list is a circular linked list.
LinkedList 17	Split a Circular linked list into two halves.
LinkedList 18	Write a Program to check whether the Singly Linked list is a palindrome or not.
LinkedList 19	Deletion from a Circular Linked List.
LinkedList 20	Reverse a Doubly Linked list.
LinkedList 21	Find pairs with a given sum in a DLL.
LinkedList 22	Count triplets in a sorted DLL whose sum is equal to given value “X”.
LinkedList 23	Sort a “k”sorted Doubly Linked list.[Very IMP]
LinkedList 24	Rotate DoublyLinked list by N nodes.
LinkedList 25	Rotate a Doubly Linked list in group of Given Size.[Very IMP]
LinkedList 26	Can we reverse a linked list in less than O(n) ?
LinkedList 27	Why Quicksort is preferred for. Arrays and Merge Sort for LinkedLists ?
LinkedList 28	Flatten a Linked List
LinkedList 29	Sort a LL of 0's, 1's and 2's
LinkedList 30	Clone a linked list with next and random pointer
LinkedList 31	Merge K sorted Linked list
LinkedList 32	Multiply 2 no. represented by LL
LinkedList 33	Delete nodes which have a greater value on right side

LinkedList 34	Segregate even and odd nodes in a Linked List
LinkedList 35	Program for n'th node from the end of a Linked List
LinkedList 36	Find the first non-repeating character from a stream of characters

DAY 78-91 in 12Th & 13Th Week

(Learn Binary Trees & Binary Search Trees)

Do First Binary Trees & Binary Search Trees
Then Start to solve these Questions..
✅ You have **13 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 13 days**, you will do later **when you practice** (during your future revision or practice sessions).

Binary Trees Question:

Binary Trees 1	level order traversal
Binary Trees 2	Reverse Level Order traversal
Binary Trees 3	Height of a tree
Binary Trees 4	Diameter of a tree
Binary Trees 5	Mirror of a tree
Binary Trees 6	Inorder Traversal of a tree both using recursion and Iteration
Binary Trees 7	Preorder Traversal of a tree both using recursion and Iteration
Binary Trees 8	Postorder Traversal of a tree both using recursion and Iteration
Binary Trees 9	Left View of a tree
Binary Trees 10	Right View of Tree
Binary Trees 11	Top View of a tree
Binary Trees 12	Bottom View of a tree
Binary Trees 13	Zig-Zag traversal of a binary tree
Binary Trees 14	Check if a tree is balanced or not
Binary Trees 15	Diagonal Traversal of a Binary tree
Binary Trees 16	Boundary traversal of a Binary tree
Binary Trees 17	Construct Binary Tree from String with Bracket Representation

Binary Trees 18	Convert Binary tree into Doubly Linked List
Binary Trees 19	Convert Binary tree into Sum tree
Binary Trees 20	Construct Binary tree from Inorder and preorder traversal
Binary Trees 21	Find minimum swaps required to convert a Binary tree into BST
Binary Trees 22	Check if Binary tree is Sum tree or not
Binary Trees 23	Check if all leaf nodes are at same level or not
Binary Trees 24	Check if a Binary Tree contains duplicate subtrees of size 2 or more [IMP]
Binary Trees 25	Check if 2 trees are mirror or not
Binary Trees 26	Sum of Nodes on the Longest path from root to leaf node
Binary Trees 27	Check if the given graph is a tree or not. [IMP]
Binary Trees 28	Find Largest subtree sum in a tree
Binary Trees 29	Maximum Sum of nodes in Binary tree such that no two are adjacent
Binary Trees 30	Print all "K" Sum paths in a Binary tree
Binary Trees 31	Find LCA in a Binary tree
Binary Trees 32	Find distance between 2 nodes in a Binary tree
Binary Trees 33	Kth Ancestor of node in a Binary tree
Binary Trees 34	Find all Duplicate subtrees in a Binary tree [IMP]
Binary Trees 35	Tree Isomorphism Problem

Binary Search Trees Question:

Binary Search Trees 1	Find a value in a BST
Binary Search Trees 2	Deletion of a node in a BST
Binary Search Trees 3	Find min and max value in a BST
Binary Search Trees 4	Find inorder successor and inorder predecessor in a BST
Binary Search Trees 5	Check if a tree is a BST or not
Binary Search Trees 6	Populate Inorder successor of all nodes
Binary Search Trees 7	Find LCA of 2 nodes in a BST
Binary Search Trees 8	Construct BST from preorder traversal
Binary Search Trees 9	Convert Binary tree into BST
Binary Search Trees 10	Convert a normal BST into a Balanced BST
Binary Search Trees 11	Merge two BST [V.V.V>IMP]
Binary Search Trees 12	Find Kth largest element in a BST
Binary Search Trees 13	Find Kth smallest element in a BST
Binary Search Trees 14	Count pairs from 2 BST whose sum is equal to given value "X"
Binary Search Trees 15	Find the median of BST in O(n) time and O(1) space
Binary Search Trees 16	Count BST nodes that lie in a given range
Binary Search Trees 17	Replace every element with the least greater element on its right
Binary Search Trees 18	Given "n" appointments, find the conflicting appointments
Binary Search Trees 19	Check preorder is valid or not
Binary Search Trees 20	Check whether BST contains Dead end
Binary Search Trees 21	Largest BST in a Binary Tree [V.V.V.V.V IMP]
Binary Search Trees 22	Flatten BST to sorted list

DAY 92-98 in 14Th Week

(Learn Recursion & Backtracking) Most important

Do First Recursion & Backtracking
Then Start to solve these Questions..
✅ You have **7 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 7 days**, you will do later **when you practice** (during your future revision or practice sessions).

BackTracking 1	Rat in a maze Problem
BackTracking 2	Printing all solutions in N-Queen Problem
BackTracking 3	Word Break Problem using Backtracking
BackTracking 4	Remove Invalid Parentheses
BackTracking 5	Sudoku Solver
BackTracking 6	m Coloring Problem
BackTracking 7	Print all palindromic partitions of a string
BackTracking 8	Subset Sum Problem
BackTracking 9	The Knight's tour problem
BackTracking 10	Tug of War
BackTracking 11	Find shortest safe route in a path with landmines
BackTracking 12	Combinational Sum
BackTracking 13	Find Maximum number possible by doing at-most K swaps
BackTracking 14	Print all permutations of a string
BackTracking 15	Find if there is a path of more than k length from a source
BackTracking 16	Longest Possible Route in a Matrix with Hurdles
BackTracking 17	Print all possible paths from top left to bottom right of a mXn matrix
BackTracking 18	Partition of a set intoK subsets with equal sum
BackTracking 19	Find the K-th Permutation Sequence of first N natural numbers

DAY 99-101 in 15Th Week

(Learn Bit Manipulation) Most important

Do First Bit Manipulation
Then Start to solve these Questions..
✅ You have **3 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 3 days**, you will do later **when you practice** (during your future revision or practice sessions).

Bit Manipulation 1	Count set bits in an integer
Bit Manipulation 2	Find the two non-repeating elements in an array of repeating elements

Bit Manipulation 3	Count number of bits to be flipped to convert A to B
Bit Manipulation 4	Count total set bits in all numbers from 1 to n
Bit Manipulation 5	Program to find whether a no is power of two
Bit Manipulation 6	Find position of the only set bit
Bit Manipulation 7	Copy set bits in a range
Bit Manipulation 8	Divide two integers without using multiplication, division and mod operator
Bit Manipulation 9	Calculate square of a number without using *, / and pow()
Bit Manipulation 10	Power Set

DAY 102-112 in 15Th & 16Th Week

(Learn Stack & Queue) Most important

Do First Stacks & Queues
Then Start to solve these Questions..
✅ You have **10 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 10 days**, you will do later **when you practice** (during your future revision or practice sessions).

Stacks & Queues 1	Implement Stack from Scratch
Stacks & Queues 2	Implement Queue from Scratch
Stacks & Queues 3	Implement 2 stack in an array
Stacks & Queues 4	find the middle element of a stack
Stacks & Queues 5	Implement "N" stacks in an Array
Stacks & Queues 6	Check the expression has valid or Balanced parenthesis or not.
Stacks & Queues 7	Reverse a String using Stack
Stacks & Queues 8	Design a Stack that supports getMin() in O(1) time and O(1) extra space.
Stacks & Queues 9	Find the next Greater element
Stacks & Queues 10	The celebrity Problem
Stacks & Queues 11	Arithmetic Expression evaluation
Stacks & Queues 12	Evaluation of Postfix expression
Stacks & Queues 13	Implement a method to insert an element at its bottom without using any other data structure.
Stacks & Queues 14	Reverse a stack using recursion

Stacks & Queues 15	Sort a Stack using recursion
Stacks & Queues 16	Merge Overlapping Intervals
Stacks & Queues 17	Largest rectangular Area in Histogram
Stacks & Queues 18	Length of the Longest Valid Substring
Stacks & Queues 19	Expression contains redundant bracket or not
Stacks & Queues 20	Implement Stack using Queue
Stacks & Queues 21	Implement Stack using Deque
Stacks & Queues 22	Stack Permutations (Check if an array is stack permutation of other)
Stacks & Queues 23	Implement Queue using Stack
Stacks & Queues 24	Implement "n" queue in an array
Stacks & Queues 25	Implement a Circular queue
Stacks & Queues 26	LRU Cache Implementations
Stacks & Queues 27	Reverse a Queue using recursion
Stacks & Queues 28	Reverse the first “K” elements of a queue
Stacks & Queues 29	Interleave the first half of the queue with second half
Stacks & Queues 30	Find the first circular tour that visits all Petrol Pumps
Stacks & Queues 31	Minimum time required to rot all oranges
Stacks & Queues 32	Distance of nearest cell having 1 in a binary matrix
Stacks & Queues 33	First negative integer in every window of size “k”
Stacks & Queues 34	Check if all levels of two trees are anagrams or not.
Stacks & Queues 35	Sum of minimum and maximum elements of all subarrays of size “k”.
Stacks & Queues 36	Minimum sum of squares of character counts in a given string after removing “k” characters.
Stacks & Queues 37	Queue based approach or first non-repeating character in a stream.
Stacks & Queues 38	Next Smaller Element

DAY 113-119 in 17Th Week

(Learn Two Pointer & Sliding Window) Most important

Do This Full Two Pointer & Sliding Window

✅ You have **7 days to complete this above playlist and practice with this playlist**.

DAY 120-126 in 18Th Week

(Learn Heap)

Do First Heap
Then Start to solve these Questions..
✅ You have **7 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 7 days**, you will do later **when you practice** (during your future revision or practice sessions).

Heap 1	Implement a Maxheap/MinHeap using arrays and recursion.
Heap 2	Sort an Array using heap. (HeapSort)
Heap 3	Maximum of all subarrays of size k.
Heap 4	"k" largest element in an array
Heap 5	Kth smallest and largest element in an unsorted array
Heap 6	Merge "K" sorted arrays. [IMP]
Heap 7	Merge 2 Binary Max Heaps
Heap 8	Kth largest sum continuous subarrays
Heap 9	Leetcode- reorganize strings
Heap 10	Merge "K" Sorted Linked Lists [V.IMP]
Heap 11	Smallest range in "K" Lists
Heap 12	Median in a stream of Integers
Heap 13	Check if a Binary Tree is Heap
Heap 14	Connect "n" ropes with minimum cost
Heap 15	Convert BST to Min Heap
Heap 16	Convert min heap to max heap

Heap 17	Rearrange characters in a string such that no two adjacent are the same.
Heap 18	Minimum sum of two numbers formed from digits of an array

DAY 127-133 in 19Th Week

(Learn Greedy Algorithm)

Do First Greedy Algorithm
Then Start to solve these Questions..

✅ You have **7 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 7 days**, you will do later **when you practice** (during your future revision or practice sessions).

Greedy 1	Activity Selection Problem
Greedy 2	Job SequencingProblem
Greedy 3	Huffman Coding
Greedy 4	Water Connection Problem
Greedy 5	Fractional Knapsack Problem
Greedy 6	Greedy Algorithm to find Minimum number of Coins
Greedy 7	Maximum trains for which stoppage can be provided
Greedy 8	Minimum Platforms Problem
Greedy 9	Buy Maximum Stocks if i stocks can be bought on i-th day
Greedy 10	Find the minimum and maximum amount to buy all N candies
Greedy 11	Minimize Cash Flow among a given set of friends who have borrowed money from each other
Greedy 12	Minimum Cost to cut a board into squares
Greedy 13	Check if it is possible to survive on Island

Greedy 14	Find maximum meetings in one room
Greedy 15	Maximum product subset of an array
Greedy 16	Maximize array sum after K negations
Greedy 17	Maximize the sum of arr[i]*i
Greedy 18	Maximum sum of absolute difference of an array
Greedy 19	Maximize sum of consecutive differences in a circular array
Greedy 20	Minimum sum of absolute difference of pairs of two arrays
Greedy 21	Program for Shortest Job First (or SJF) CPU Scheduling
Greedy 22	Program for Least Recently Used (LRU) Page Replacement algorithm
Greedy 23	Smallest subset with sum greater than all other elements
Greedy 24	Chocolate Distribution Problem
Greedy 25	DEFKIN -Defense of a Kingdom
Greedy 26	DIEHARD -DIE HARD
Greedy 27	GERGOVIA -Wine trading in Gergovia
Greedy 28	Picking Up Chicks
Greedy 29	CHOCOLA –Chocolate
Greedy 30	ARRANGE -Arranging Amplifiers
Greedy 31	K Centers Problem
Greedy 32	Minimum Cost of ropes
Greedy 33	Find smallest number with given number of digits and sum of digits
Greedy 34	Rearrange characters in a string such that no two adjacent are same
Greedy 35	Find maximum sum possible equal sum of three stacks

DAY 134-147 in 20Th & 21Th Week

(Learn Graph)

Do First Graph
Then Start to solve these Questions..

✅ You have **14 days to complete as many questions as possible**.

✅ **Any questions you don't complete in these 14 days**, you will do later **when you practice** (during your future revision or practice sessions).

Graph 1	Create a Graph, print it
Graph 2	Implement BFS algorithm
Graph 3	Implement DFS Algo
Graph 4	Detect Cycle in Directed Graph using BFS/DFS Algo
Graph 5	Detect Cycle in UnDirected Graph using BFS/DFS Algo
Graph 6	Search in a Maze
Graph 7	Minimum Step by Knight
Graph 8	flood fill algo
Graph 9	Clone a graph
Graph 10	Making wired Connections
Graph 11	word Ladder
Graph 12	Dijkstra algo
Graph 13	Implement Topological Sort
Graph 14	Minimum time taken by each job to be completed given by a Directed Acyclic Graph
Graph 15	Find whether it is possible to finish all tasks or not from given dependencies
Graph 16	Find the no. of Islands
Graph 17	Given a sorted Dictionary of an Alien Language, find order of characters
Graph 18	Implement Kruksal'sAlgorithm
Graph 19	Implement Prim's Algorithm
Graph 20	Total no. of Spanning tree in a graph
Graph 21	Implement Bellman Ford Algorithm
Graph 22	Implement Floyd warshallAlgorithm
Graph 23	Travelling Salesman Problem
Graph 24	Graph ColouringProblem
Graph 25	Snake and Ladders Problem
Graph 26	Find bridge in a graph
Graph 27	Count Strongly connected Components(Kosaraju Algo)
Graph 28	Check whether a graph is Bipartite or Not
Graph 29	Detect Negative cycle in a graph

Graph 30	Longest path in a Directed Acyclic Graph
Graph 31	Journey to the Moon
Graph 32	Cheapest Flights Within K Stops
Graph 33	Oliver and the Game
Graph 34	Water Jug problem using BFS
Graph 35	Water Jug problem using BFS
Graph 36	Find if there is a path of more thank length from a source
Graph 37	M-ColouringProblem
Graph 38	Minimum edges to reverse o make path from source to destination
Graph 39	Paths to travel each nodes using each edge(Seven Bridges)
Graph 40	Vertex Cover Problem
Graph 41	Chinese Postman or Route Inspection
Graph 42	Number of Triangles in a Directed and Undirected Graph
Graph 43	Minimise the cashflow among a given set of friends who have borrowed money from each other
Graph 44	Two Clique Problem

DAY 148-152 in 22nd Week

(Learn Tries)

Do First Tries
Then Start to solve these Questions..

✅ You have **5 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 5 days**, you will do later **when you practice** (during your future revision or practice sessions).

Trie 1	Construct a trie from scratch
Trie 2	Find shortest unique prefix for every word in a given list
Trie 3	Word Break Problem | (Trie solution)
Trie 4	Given a sequence of words, print all anagrams together
Trie 5	Implement a Phone Directory
Trie 6	Print unique rows in a given boolean matrix

DAY 153-168 in 23rd & 24Th Week

(Learn Dynamic Programming)

Do First Dynamic Programming
Then Start to solve these Questions..

✅ You have **16 days to complete as many questions as possible**.
✅ **Any questions you don't complete in these 16 days**, you will do later **when you practice** (during your future revision or practice sessions).

Dynamic Programming 1	Coin ChangeProblem
Dynamic Programming 2	Knapsack Problem
Dynamic Programming 3	Binomial CoefficientProblem
Dynamic Programming 4	Permutation CoefficientProblem
Dynamic Programming 5	Program for nth Catalan Number
Dynamic Programming 6	Matrix Chain Multiplication
Dynamic Programming 7	Edit Distance
Dynamic Programming 8	Subset Sum Problem
Dynamic Programming 9	Friends Pairing Problem
Dynamic Programming 10	Gold Mine Problem
Dynamic Programming 11	Assembly Line SchedulingProblem
Dynamic Programming 12	Painting the Fence problem
Dynamic Programming 13	Maximize The Cut Segments
Dynamic Programming 14	Longest Common Subsequence

Dynamic Programming 15	Longest Repeated Subsequence
Dynamic Programming 16	Longest Increasing Subsequence
Dynamic Programming 17	Space Optimized Solution of LCS
Dynamic Programming 18	LCS (Longest Common Subsequence) of three strings
Dynamic Programming 19	Maximum Sum Increasing Subsequence
Dynamic Programming 20	Count all subsequences having product less than K
Dynamic Programming 21	Longest subsequence such that difference between adjacent is one
Dynamic Programming 22	Maximum subsequence sum such that no three are consecutive
Dynamic Programming 23	Egg Dropping Problem
Dynamic Programming 24	Maximum Length Chain of Pairs
Dynamic Programming 25	Maximum size square sub-matrix with all 1s
Dynamic Programming 26	Maximum sum of pairs with specific difference
Dynamic Programming 27	Min Cost PathProblem
Dynamic Programming 28	Maximum difference of zeros and ones in binary string
Dynamic Programming 29	Minimum number of jumps to reach end
Dynamic Programming 30	Minimum cost to fill given weight in a bag
Dynamic Programming 31	Minimum removals from array to make max –min <= K
Dynamic Programming 32	Longest Common Substring
Dynamic Programming 33	Count number of ways to reacha given score in a game
Dynamic Programming 34	Count Balanced Binary Trees of Height h
Dynamic Programming 35	LargestSum Contiguous Subarray [V>V>V>V IMP]
Dynamic Programming 36	Smallest sum contiguous subarray
Dynamic Programming 37	Unbounded Knapsack (Repetition of items allowed)
Dynamic Programming 38	Word Break Problem
Dynamic Programming 39	Largest Independent Set Problem
Dynamic Programming 40	Partition problem
Dynamic Programming 41	Longest Palindromic Subsequence
Dynamic Programming 42	Count All Palindromic Subsequence in a given String
Dynamic Programming 43	Longest Palindromic Substring

Dynamic Programming 44	Longest alternating subsequence
Dynamic Programming 45	Weighted Job Scheduling
Dynamic Programming 46	Coin game winner where every player has three choices
Dynamic Programming 47	Count Derangements (Permutation such that no element appears in its original position) [IMPORTANT]
Dynamic Programming 48	Maximum profit by buying and selling a share at most twice [IMP]
Dynamic Programming 49	Optimal Strategy for a Game
Dynamic Programming 50	Optimal Binary Search Tree
Dynamic Programming 51	Palindrome PartitioningProblem
Dynamic Programming 52	Word Wrap Problem
Dynamic Programming 53	Mobile Numeric Keypad Problem [IMP]
Dynamic Programming 54	Boolean Parenthesization Problem
Dynamic Programming 55	Largest rectangular sub-matrix whose sum is 0
Dynamic Programming 56	Largest area rectangular sub-matrix with equal number of 1's and 0's [IMP]
Dynamic Programming 57	Maximum sum rectangle in a 2D matrix
Dynamic Programming 58	Maximum profit by buying and selling a share at most k times
Dynamic Programming 59	Find if a string is interleaved of two other strings
Dynamic Programming 60	Maximum Length of Pair Chain

✅ **We completed the DSA syllabus in 6 months.**
But many of you couldn't solve all the questions within the expected time.
📢 **Now is the time to catch up.**
Start solving those questions — one by one, with full focus.

⚔️ Also participate in regular contests & coding hackathons.

- 🔗 Leetcode Weekly Contest

- 🔗 [Codeforces Contests]
- 🔗[Unstop Hackathons]
- 🔗 [Devfolio Hackathons]

✅ **Hope you've completed all the DSA questions so far.**
Now, we're giving you the **most asked coding questions** and a special **Placement Sheet**.

🎯 **Before you attend your first interview, make sure to solve every question in this sheet.**
This could be the difference between rejection and selection.

Most Asked Coding Questions In Interview

#	Question Title	Topic	Difficulty	Link
1	Sum of Dependencies in a Graph	Graphs	Medium	(GeeksforGeeks)
2	Activity Selection Problem	Greedy	Medium	(GeeksforGeeks)
3	Interesting Facts about Bitwise Operators in Java	Bit Manipulation	Medium	(GeeksforGeeks)
4	Sliding Window Maximum	Arrays/Deque	Hard	(Leetcode)
5	Count Inversions in an Array (Merge Sort)	Arrays/Divide & Conquer	Hard	(GeeksforGeeks)
6	Chocolate Distribution Problem	Arrays/Greedy	Medium	(GeeksforGeeks)
7	Print All Possible Words from Phone Digits	Recursion/Backtracking	Medium	(GeeksforGeeks)
8	Largest Sum Contiguous Subarray (Kadane's Algorithm)	Arrays	Medium	(GeeksforGeeks)
9	Find the Middle of a Given Linked List	Linked List	Medium	(GeeksforGeeks)
10	Level Order Tree Traversal	Trees	Medium	(GeeksforGeeks)
11	Reverse Level Order Traversal	Trees	Medium	(GeeksforGeeks)
12	Level Order Traversal in Spiral Form	Trees	Medium	(GeeksforGeeks)
13	Print BST Keys in the Given Range	BST	Medium	(GeeksforGeeks)
14	Evaluation of Expression Tree	Trees/Stack	Medium	(GeeksforGeeks)
15	Implement Queue using Stacks	Stack/Queue	Medium	(GeeksforGeeks)

16	Maximum Width of a Binary Tree	Trees	Medium	(GeeksforGeeks)
17	Kadane's Algorithm	Arrays	Medium	(GeeksforGeeks)
18	Key Pair (Check if Array has Pair with Given Sum)	Arrays/Hashing	Medium	(GeeksforGeeks)
19	Search in a Rotated Array	Arrays/Binary Search	Medium	(GeeksforGeeks)
20	0-1 Knapsack Problem	Dynamic Programming	Medium	(GeeksforGeeks)
21	Stock Buy and Sell	Arrays	Medium	(GeeksforGeeks)
22	Left View of Binary Tree	Trees	Medium	(GeeksforGeeks)
23	Add Two Numbers Represented by Linked Lists	Linked List	Medium	(GeeksforGeeks)
24	Get Minimum Element from Stack	Stack	Medium	(GeeksforGeeks)
25	Merge Two Sorted Linked Lists	Linked List	Medium	(GeeksforGeeks)
26	Merge k Sorted Linked Lists	Linked List/Heap	Hard	(LeetCode)
27	Clone a Linked List with Random Pointers	Linked List	Hard	(GeeksforGeeks)
28	Intersection Point of Two Linked Lists	Linked List	Medium	(GeeksforGeeks)
29	Reverse a Linked List in Groups of Given Size	Linked List	Medium	(GeeksforGeeks)
30	Detect Cycle in a Directed Graph	Graphs	Medium	(GeeksforGeeks)
31	Topological Sorting	Graphs	Medium	(GeeksforGeeks)
32	Dijkstra’s Shortest Path Algorithm	Graphs	Medium	(GeeksforGeeks)
33	Word Break Problem	Dynamic Programming	Medium	(GeeksforGeeks)
34	Longest Increasing Subsequence	Dynamic Programming	Medium	(GeeksforGeeks)
35	Longest Common Subsequence	Dynamic Programming	Medium	(GeeksforGeeks)
36	Unique Paths II	Dynamic Programming	Medium	(LeetCode)
37	Minimum Window Substring	Strings/Sliding Window	Hard	(LeetCode)
38	Trapping Rain Water	Arrays/Stack	Hard	(LeetCode)
39	Median of Two Sorted Arrays	Arrays/Binary Search	Hard	(LeetCode)
40	3Sum	Arrays/Two Pointers	Medium	(LeetCode)
41	Longest Substring Without Repeating Characters	Strings/Hashing	Medium	(LeetCode)
42	Merge Intervals	Arrays/Sorting	Medium	(LeetCode)

43	Valid Parentheses	Stack	Medium	(LeetCode)
44	Binary Tree Level Order Traversal	Trees	Medium	(LeetCode)
45	Word Search	Backtracking	Medium	(LeetCode)
46	Unique Paths	Dynamic Programming	Medium	(LeetCode)
47	Subsets	Backtracking	Medium	(LeetCode)
48	Find All Anagrams in a String	Strings/Sliding Window	Medium	(LeetCode)
49	Course Schedule (Cycle Detection in Graph)	Graphs	Medium	(LeetCode)
50	Minimum Number of Platforms Required for a Railway/Bus Station	Greedy/Sorting	Medium	(GeeksforGeeks)
51	Number of Connected Components in Graph	Graphs / Union-Find	Medium	(GeeksforGeeks)
52	Implement Trie (Prefix Tree)	Trie / Strings	Medium	(LeetCode)
53	Rotate Matrix by 90 Degrees	Matrix Traversal	Medium	(LeetCode)
54	Spiral Matrix	Matrix Traversal	Medium	(LeetCode)
55	Search a 2D Matrix	Arrays / Binary Search	Medium	(LeetCode)
56	N-Queens Problem	Backtracking	Hard	(LeetCode)
57	Sudoku Solver	Backtracking	Hard	(LeetCode)
58	Minimum Path Sum	Dynamic Programming	Medium	(LeetCode)
59	Coin Change	Dynamic Programming	Medium	(LeetCode)
60	Combination Sum	Backtracking	Medium	(LeetCode)